Data Preprocessing, Normalization, and Normalization Strategies for EV Uptake Proteomics

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Clarification of TMT Channel Design and Terminology

The TMT plex channels were assigned as follows:

- **Boost Channels (126 & 127N):**

Formed by concentrated EV material to increase precursor ion currents for MS2 triggering and RTS selection. These channels were excluded from all quantitative uptake calculations.

- **Spike-in Channels (133N — 134C):**

Formed by a synthetic mixture of lysates of recipient cell (SAOS2) combined with EVs (either PC3 or PC3M). The term “Spike-in” designates this reference standard used for per-protein ratio normalization, and does not refer to the boost channels.

- **Uptake Channels (130N — 132C & 135N):**

Formed by biological replicates of recipient SAOS2 cells incubated with PC3 EV or PC3M EV preparations.

TMT channel	Sample content
126	PC3 Evs
127N	PC3M Evs
127C	*empty*
128N	*empty*
128C	SAOS2
129N	SAOS2
129C	SAOS2
130N	SAOS2 + PC3 EV
130C	SAOS2 + PC3 EV
131N	SAOS2 + PC3 EV
131C	SAOS2 + PC3M EV
132N	SAOS2 + PC3M EV
132C	SAOS2 + PC3M EV
133N	SAOS2 + spike PC3 EV
133C	SAOS2 + spike PC3 EV
134N	SAOS2 + spike PC3M EV
134C	SAOS2 + spike PC3M EV
135N	SAOS2 + PC3M EV

1. Data Curation and Peptide Preprocessing Workflow

Quantitative analysis was performed on *MEK_25_02_label_included.txt*. This file combines all fractions into a single dataset with 88,138 peptide identifications and covers the full set of TMT channels, controls, and spike-in standards. Datasets from individual fractions and alternative instrument settings were set aside to ensure a uniform processing workflow focused on biological uptake differences.

To evaluate dataset consistency prior to quantification, entries were inspected across four baseline metrics: metabolic labeling, external contaminants, peptide uniqueness, and identification false discovery rate (FDR) (Table 1, Figure 1). All 88,138 records passed search-engine filtering with $q \leq 0.01$ (FDR 1%), with zero entries flagged as common laboratory contaminants (Figure 1B, D). Light peptides from the recipient SAOS-2 cellular

background accounted for 97.5% of entries (n = 85,929), while heavy-labeled peptides from vesicular cargo accounted for 2.5% (n = 2,209) (Figure 1A). Furthermore, 97.7% of the sequences mapped uniquely to a single master protein accession (Figure 1C).

Table 1. Baseline Filtering Summary. *Delineates the input dataset structure at the peptide level prior to abundance filtering.*

QC Metric	Value / Criterion	Peptide Count (n)	Share of Pool (%)
Total Identifications	Raw output	88,138	100.0%
PSM Spectral Confidence	q-value <= 0.01 (FDR 1%)	88,138	100.0%
Contaminants	Contaminant == False	88,138	100.0%
Mapping Specificity	Number of Proteins == 1	86,106	97.7%
	Number of Proteins > 1	2,032	2.3%
Metabolic Origin	Label == 'light' (recipient cells)	85,929	97.5%
	Label == 'heavy' (EV cargo)	2,209	2.5%

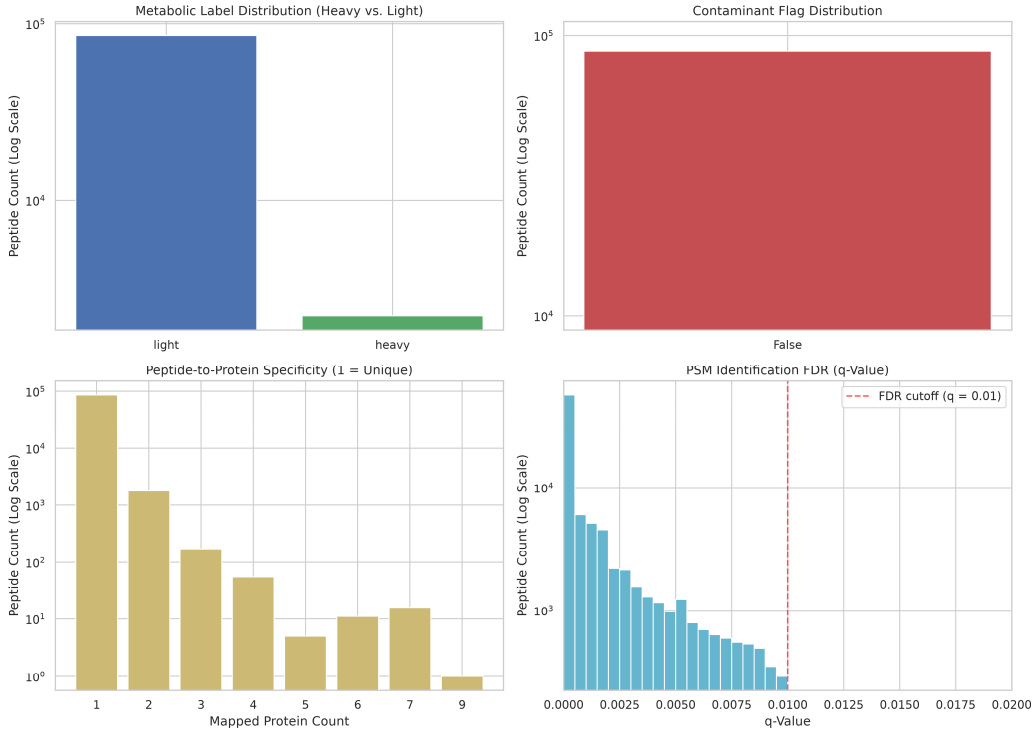


Figure 1. Initial spectral library curation, identification quality control, and peptide specificity*. (A) Distribution of metabolic labels across all identified peptide-spectrum matches (PSMs; total N = 88,138). The background endogenous recipient cell proteome (light label) accounts for 97.5% (n = 85,929), whereas heavy-labeled EV-derived peptides represent 2.5% (n = 2,209) of the total population. (B) Evaluation of external laboratory artifacts against the standard contaminant database; 100% of the identifications (n = 88,138) were confirmed as non-contaminant protein sequences. (C) Peptide-to-protein mapping specificity. A total of 86,106 peptides (97.7%) mapped uniquely to a single master protein accession (mapped count = 1), while 2,032 peptides mapped to multiple related protein group entries. (D) False discovery rate (FDR) control for PSM identifications. All accepted entries satisfied the stringent threshold of q-value ≤ 0.01 (100.0%, median q-value = 1.05×10^{-4}), with the distribution heavily skewed toward high-confidence matches (q-value < 0.001), well below the nominal 1% FDR boundary (red dashed line).*

Enzymatic cleavage was evaluated by calculating sequence length distributions for all identified peptides (Figure 2). Sequence lengths ranged from 7 to 30 amino acids, with a median of 12 amino acids and a mean of 13.13 amino acids (Figure 2A). Comparing the density curves of the light cellular background (n = 85,929) and heavy vesicular cargo (n = 2,209) showed matching distributions (Figure 2B), confirming consistent tryptic digestion across both pools without size-dependent recovery artifacts.

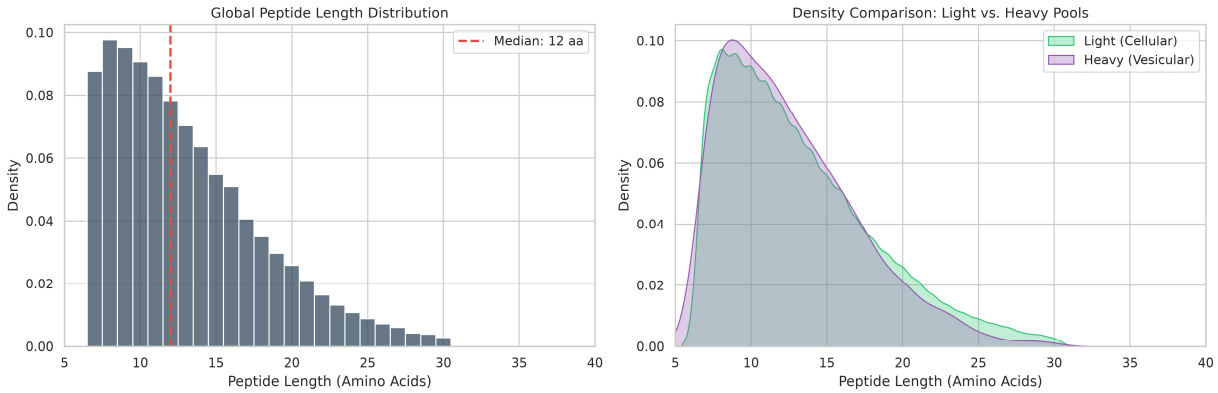


Figure 2. Peptide sequence length distribution across identified pools. *(A) Global length distribution for all identified peptides (N = 88,138). Peptide lengths range from 7 to 30 amino acids (mean = 13.13 aa), with a median of 12 amino acids (red dashed line), consistent with expected tryptic digestion profiles. (B) Kernel density estimation comparing sequence lengths of light (endogenous cellular, n = 85,929) and heavy (vesicular cargo, n = 2,209) peptide pools. Both pools exhibit overlapping length distributions, indicating no enzymatic digestion bias between recipient cellular proteins and internalized vesicle-derived cargo.*

Channel occupancy was calculated across all 18 multiplex channels (Figure 3). The overall distribution showed a median of 0.0 active channels across both light and heavy pools. For light peptides, 97.7% (n = 83,963) contained zero active reporter channels, reflecting that MS3 triggering was targeted toward heavy labeled precursors. The remaining 2.3% of light peptides (n = 1,966) showing non-zero reporter intensity stem from precursor co-isolation within the quadrupole selection window during multiplexed acquisition. For heavy peptides, 1,018 of 2,209 features showed non-zero signal, forming a distribution centered near 14 active channels (Figure 3).

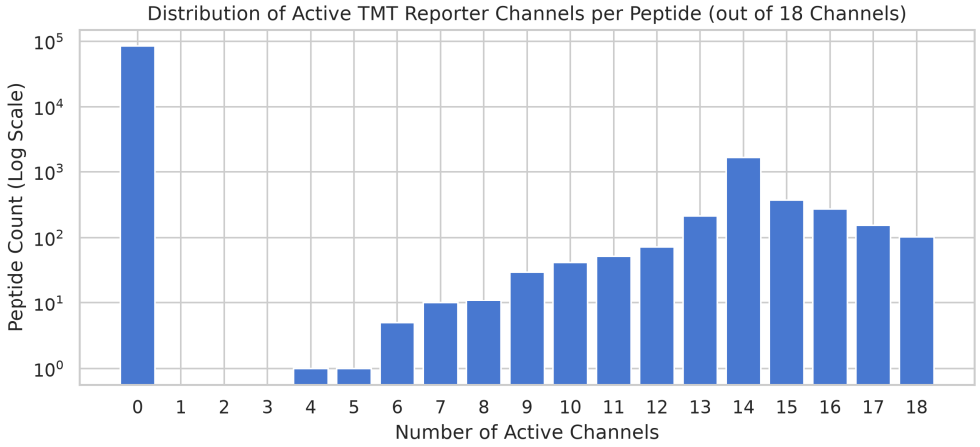


Figure 3. TMT reporter channel occupancy per peptide across the multiplex. *Distribution of active reporter ion channels (channels with detected intensity out of 18 total multiplexed channels) across all identified peptides (N = 88,138). The vast majority of peptides exhibit zero active channels (n = 85,551), corresponding primarily to unquantified light background peptides (mean active channels = 0.32, count = 85,929). Peptides with quantifiable signal display a distinct distribution centered around 14 active channels, driven predominantly by the heavy-labeled vesicular peptide pool (mean active channels = 6.63, count = 2,209). Peptides lacking reporter signal across all channels were filtered out prior to protein-level rollup.*

To explain the presence of heavy peptides lacking reporter signals, the diagnostic flag Quan Info was evaluated (Table 2). The primary cause of missing values was instrument duty-cycle constraints during real-time search (RTS): 53.2% of heavy peptides (n = 1,175) were identified at the MS2 level but were not selected for an MS3 scan (NoQuanValues). Only 0.7% (n = 16) were acquired and subsequently discarded due to spectral isolation interference (ExcludedByMethod). The remaining 1,018 peptides (46.1%) produced valid reporter ion quantifications.

Table 2. Investigation of Unquantified Heavy Peptides (Quan Info).

Status Flag	Peptide Count (n)	Share of Heavy (%)	Analytical Cause
NaN (Passed)	1018	46.1%	Valid SPS-MS3 scan, passed all filtering criteria.
NoQuanValues	1175	53.2%	Precursor confirmed at MS2, but MS3 was not triggered (RTS duty cycle limitation).
ExcludedByMethod	16	0.7%	MS3 scan acquired but rejected due to high co-isolation or low SPS match.

To prevent carrier-driven false positives, the 1,018 heavy peptides were checked across all biological uptake channels (Channels 130N–132C, 135N). Exactly 7 peptides displayed reporter signal exclusively within the EV boost/carrier channels (Channels 126, 127N) or spike-in reference channels (Channels 133N–134C), with zero signal in target cell co-culture channels. These 7 non-informative peptides were excluded, yielding a final set of 1,011 heavy peptides for protein-level rollup.

Among the 73 ambiguous peptides, manual reassignment revealed three categories: alternative splicing isoforms of a single gene (*PKM*, *HNRNPC*), gene duplications or pseudogenes sharing identical mature sequences (*ANXA2*, core histones such as *H2AC11* and *H3-3A*), and highly conserved paralogous gene families (*ATP1A1*, *ACTB*/*ACTC1*). Manual curation and gene-level collapsing were not performed at this stage to avoid introducing arbitrary assignment bias, all multi-mapping features were set aside, and downstream processing was restricted strictly to peptides computationally mapped as unique to a single master protein accession.

2. Background Gating and Noise Floor Filtering

To distinguish genuine extracellular vesicle cargo proteins from non-specific background signals, instrument noise, and recipient cell proteomic background, a dual-threshold empirical filtering strategy was applied.

Due to reporter ion cross-talk, isotopic impurities, and baseline instrumental electronic noise typical for multiplexed SPS-MS3 workflows, negative control channels can display non-zero low-intensity readings even in the absence of target proteins. Furthermore, raw intensity distributions exhibit pronounced right-skewness and left-censoring at the instrument's physical limit of detection ($LOD \approx 1.7$), rendering standard parametric normal models inapplicable. Therefore, robust non-parametric metrics were utilized. The filtering framework relies on two complementary criteria:

1. **Absolute Noise Floor Threshold (α):**

The baseline noise profile was empirically estimated using non-zero reporter intensities across negative biological controls (recipient cells (SAOS-2), n = 3) and empty instrument channels (n = 2). Across these background channels, the median signal was found to be 6.55 in SAOS-2 controls and 7.00 in empty channels (with a 75th percentile of 12.90 and 14.90, respectively). Given the near-identical intensity distributions between empty and recipient control channels, this signal represents non-specific baseline noise. Accordingly, the

absolute threshold was set to the empirical background median ($\alpha = 6.6$). A protein was required to exceed this noise floor in at least one experimental condition:

$$\max(\text{Median}_{\text{PC3}}, \text{Median}_{\text{PC3M}}) > \alpha$$

2. **Signal-to-Background Ratio (β):**

To account for technical variability and avoid false-positive assignments due to analytical fluctuations, the required fold-change threshold (β) was derived from experimental reproducibility. The median coefficient of variation ($CV = \text{SD}/\text{Mean}$) across the replicates of the recipient control channels was determined to be 24%. Under the standard law of error propagation, the relative technical uncertainty of the ratio between two independent noisy measurements (sample vs. control) is:

$$CV_{\text{ratio}} \approx \sqrt{CV_{\text{sample}}^2 + CV_{\text{control}}^2} = \sqrt{0.24^2 + 0.24^2} \approx 0.34\ (34\%)$$

To achieve statistical confidence at the 3σ level ($p < 0.003$), an authentic cargo signal must exceed random analytical fluctuations by at least three standard deviations of the ratio ($3 \times 34\% \approx 102\%$, representing a >2-fold increase). Thus, the fold-change threshold was established at $\beta = 2.0$ ($\log_2 FC \geq 1.0$):

$$\max(\text{Median}_{\text{PC3}}, \text{Median}_{\text{PC3M}}) \geq \beta \times \text{Median}_{\text{SAOS2 control}}$$

From an initial pool of 372 strictly quantified proteins, 281 proteins (75.5%) passed the absolute baseline noise threshold ($\alpha = 6.6$), successfully filtering out near-LOD artifacts and low-level instrument fluctuations. Subsequently, applying the dual-criteria filter ($\alpha = 6.6$ and $\beta = 2.0$) retained a cohort of 212 validated EV cargo proteins (55.6% of the initial cohort), while discarding 74 proteins as non-specific background or ambiguous signals within the experimental noise margin.

Prior to downstream comparative analysis, sample loading and channel-wise quantification metrics were evaluated across all multiplexed channels. Channel *PC3_EV_R1* exhibited severe analytical suppression indicative of technical underlabeling (library preparation failure): total reporter ion intensity was reduced by more than 3.5-fold compared to corresponding biological replicates (2266.9 vs. 7448.2 and 8614.1 in PC3_EV_R2 and R3, respectively), with channel median intensity dropping to baseline noise levels (2.90) and zero-value dropouts increasing to 33.1% (compared to ≤ 7.0% in parallel replicates). Based on the aforementioned technical anomalies, PC3_EV_R1 was classified as an analytical outlier and excluded from downstream quantitative comparisons to prevent systematic dilution bias. Subsequent analysis of the 212 cargo proteins was performed across validated replicates (n=2 for PC3-EV, n=4 for PC3M-EV), yielding full data completeness without requiring synthetic imputation.

3. Internal Spike-in Normalization and Quantification of Cargo Uptake Efficiency

Raw reporter ion intensities obtained from multiplexed isobaric labeling permit direct relative comparisons of the same protein across different experimental channels. However, these raw intensities cannot be used to compare different proteins with one another within the same sample. In shotgun mass spectrometry, absolute signal abundance is governed not only by molecular concentration but also by protein-specific physicochemical determinants, including sequence length, tryptic digestion efficiency, electrospray ionization response, baseline abundance in the parent EVs, etc. Consequently, a high raw intensity can reflect either efficient cellular internalization of a minor cargo or, conversely, weak internalization of an abundant, highly flyable protein species.

To overcome this confounding effect and evaluate genuine cargo uptake efficiency, sample reporter intensities were normalized against matched spike-in channels. These spike-in channels consist of a defined mixture of recipient cell lysate and donor EVs, capturing the baseline abundance, peptide yield, and ionization characteristics of each protein species prior to biological co-culture. Calculating the protein-wise ratio of the co-culture sample intensity to the reference spike-in intensity transforms raw ion counts into a dimensionless relative uptake metric:

$$\text{Relative Uptake}_i = \frac{I_{\text{Sample},i}}{I_{\text{Spike-in},i}}$$

This ratio decouples the measured biological uptake from intrinsic baseline abundance and technical MS responsiveness. By anchoring each protein to its own input standard, the relative degree of internalization becomes directly comparable across different protein species within the internalized vesicle payload.

To account for baseline particle dosing discrepancies between vesicle preparations, an empirical $2\times$ scaling factor ($\Delta\log_2$ = -1.0) was nominally introduced for PC3 channels based on Nanoparticle Tracking Analysis (NTA) estimates. Crucially, empirical particle enumeration naturally carries inherent analytical variance and provides an order-of-magnitude approximation rather than an absolute physical constant. However, because our downstream analytical framework strictly employs sample-median centering prior to all unsupervised ordination and differential hypothesis testing, any uniform additive scalar offset in $\log_2$-space is algebraically canceled out:

$$(y_{ij} - c) - \text{median}_j(y_j - c) = y_{ij} - \text{median}_j(y_j)$$

Consequently, downstream evaluations of relative cargo selectivity and statistical significance are completely invariant to the chosen NTA scaling parameter, rendering post-hoc parameter optimization or sensitivity tuning unnecessary.

For multi-replicate reference standards, the baseline denominator was calculated as the geometric mean of raw intensities across corresponding spike-in channels, representing the arithmetic mean in $\log_2$-space. Log-transformation was performed strictly on genuine non-zero abundance measurements without the addition of arbitrary pseudocounts (such as $\log_2(I + 1)$). In isobaric labeling data operating near the lower dynamic range, standard pseudocount transformations introduce systematic non-linear compression and distort true fold-change ratios between high- and low-abundance proteins. Dropouts (0.15% of all data; 2 out of 1272 values) occurring at the instrumental detection limit were handled via conservative limit-of-detection substitutions (LOD/2) prior to transformation, preserving ratio linearity across the quantified cohort.

4. Exploratory Ordination and Resolution of Global Bulk Uptake Effects

Unsupervised Principal Component Analysis and pairwise Pearson correlation was performed on the spike-normalized, log-transformed intensity matrix across the authenticated cohort (N = 212 proteins, 6 channels). Technical and biological concordance was verified within each experimental arm. Non-metastatic PC3 replicates displayed strong inter-replicate correlation (r = 0.917), while metastatic PC3M replicates formed an internally consistent cluster (pairwise r = 0.76 - 0.94) clearly segregated from the PC3 condition (r = 0.33 - 0.57).

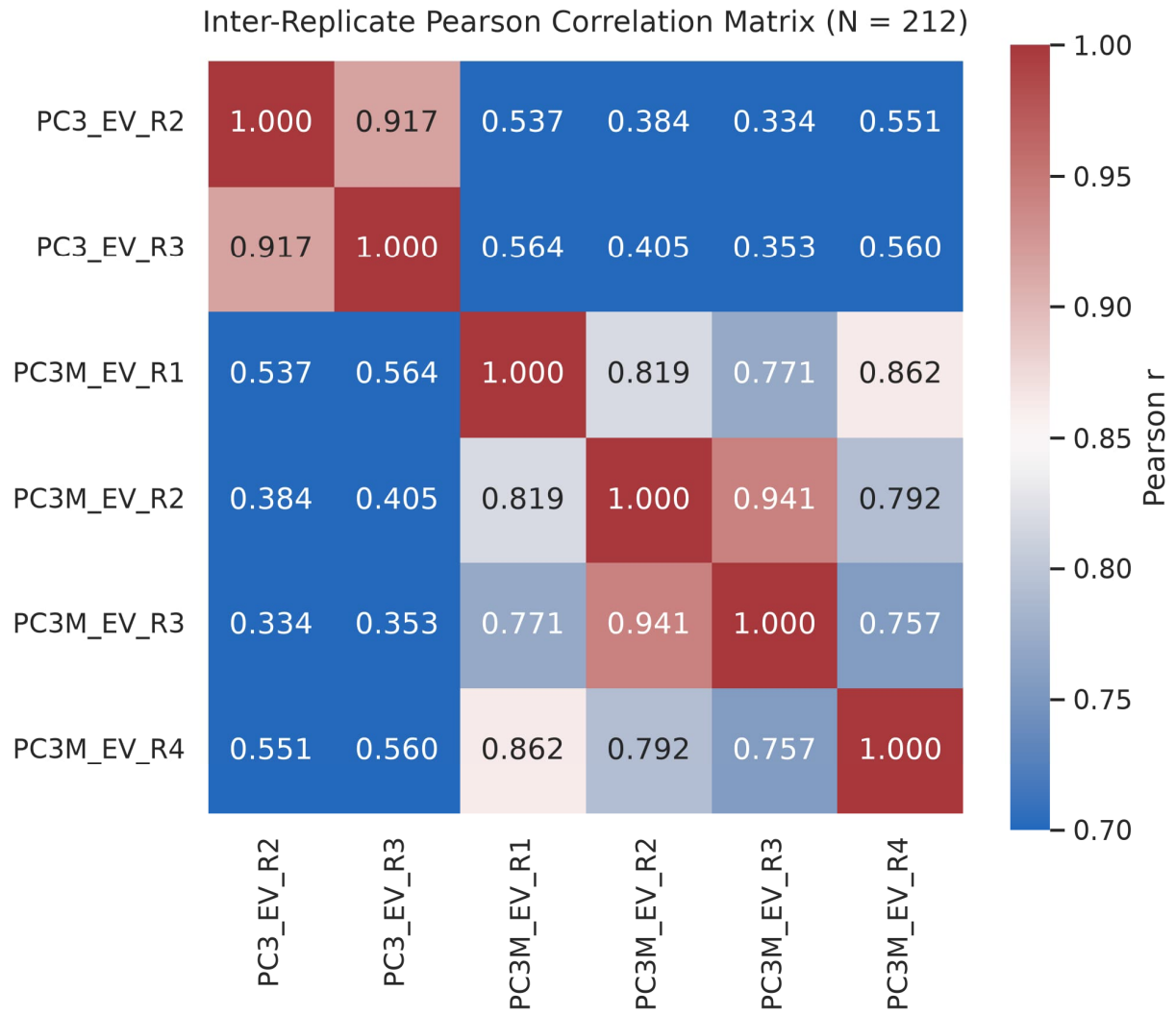


Figure 4. Inter-replicate Pearson cross-correlation matrix of normalized cargo uptake profiles. *Heatmap showing pairwise Pearson correlation coefficients computed across all quantified cargo proteins (N = 212) for the 6 validated biological channels. High intra-group consistency is observed within both non-metastatic and metastatic conditions, contrasted by distinct separation between the two biological groups.*

The ordination projection revealed that the first principal component (PC1) captured the dominant fraction of overall dataset variance (87.4%), driving a complete and pronounced separation between recipient osteoblasts treated with non-metastatic PC3 versus metastatic PC3M vesicles along the horizontal axis. The second principal component (PC2) accounted for 5.8% of total variance, reflecting residual intra-group dispersion among PC3M replicates.

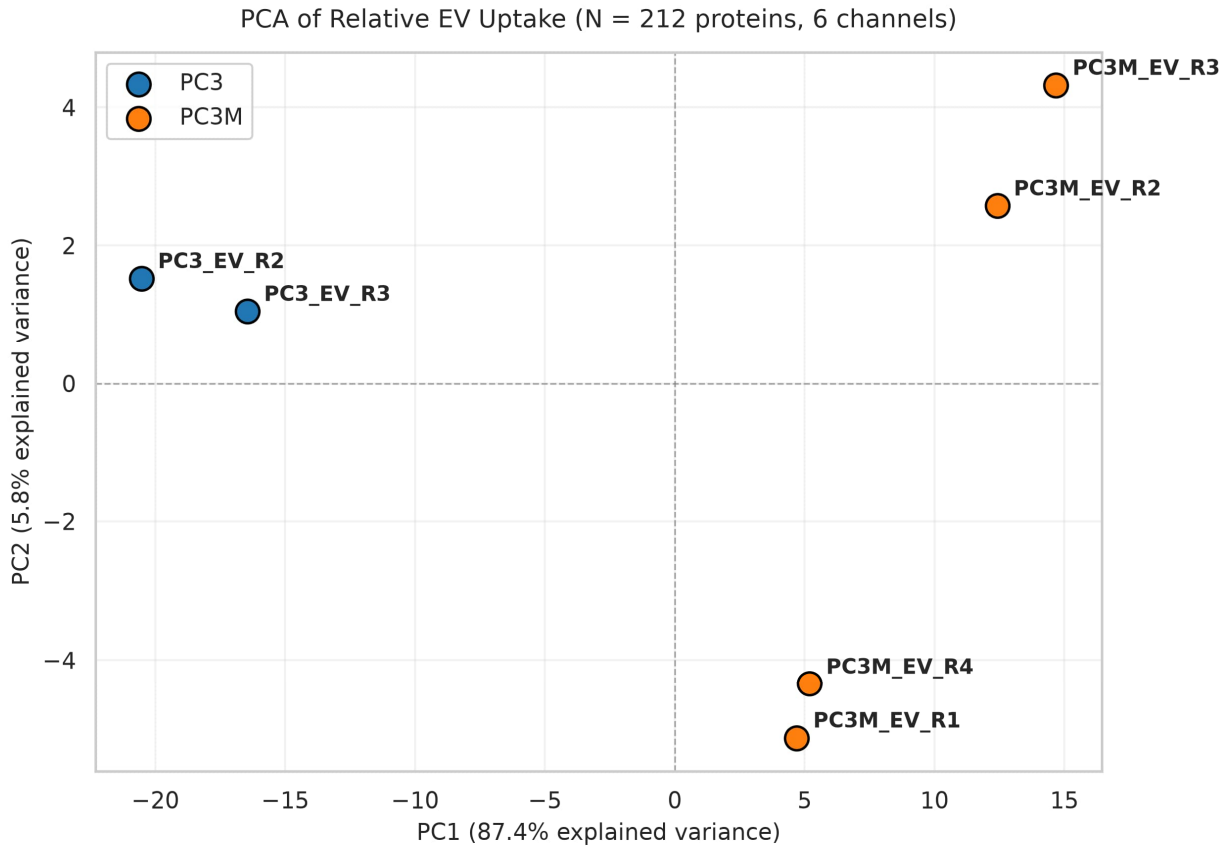


Figure 5. Principal component analysis of relative EV cargo uptake across validated biological replicates. *Score plot of unsupervised principal component analysis performed on the N = 212 spike-normalized, log_2-transformed EV cargo proteins across 6 experimental channels. PC1 accounts for 87.4% of the total variance, driving a pronounced baseline separation between recipient osteoblasts (SAOS-2) co-cultured with non-metastatic PC3-derived EVs (n = 2, blue) and metastatic PC3M-derived EVs (n = 4, orange). PC2 accounts for 5.8% of the variance, reflecting secondary dispersion across biological replicates.*

Inspection of the PC1 feature eigenvectors (loadings) revealed a pervasive directional bias across the cargo proteome: 211 out of 212 evaluated proteins (99.5%) exhibited strictly positive loading coefficients (ranging up to +0.0734), with only a single protein projecting negatively (-0.0628). In multivariate linear decomposition, an axis accounting for the vast majority of dataset variance driven by near-unanimous unidirectional loadings represents a canonical global scale effect. Rather than reflecting reciprocal pathway reprogramming or stoichiometric substitution of specific cargo molecules, PC1 captured a macro-scale biological phenomenon: recipient SAOS-2 osteoblasts internalize metastatic PC3M-derived vesicle cargo at a systematically higher absolute rate and volume than non-metastatic PC3 cargo.

While the global scale effect highlights an elevated capacity of metastatic vesicles for bulk endocytosis, this uniform elevation obscures the detection of authentic stoichiometric selectivity. Because absolute cargo uptake reflects the product of total internalized vesicle volume and individual intra-vesicular protein concentrations, non-specific bulk uptake causes nearly all proteins to appear elevated in the metastatic condition. To identify specific molecular drivers responsible for phenotypic tropism rather than passive co-internalization, this global volume component must be decoupled prior to downstream differential testing.

5. Sample-Median Centering and Resolution of Relative Cargo Selectivity

To eliminate the confounding influence of bulk endocytosis and isolate selective cargo uptake, sample-median centering was applied across all 6 authenticated channels. For each channel j, the channel-wide median across

all N = 212 quantified proteins were subtracted from each individual protein's $\log_2$-transformed normalized uptake value:

$$\text{Signal}_{\text{rel,ij}} = \text{Signal}_{\text{ij}} - \text{Median}_{\text{j}}$$

Operating in logarithmic space makes this transformation mathematically equivalent to scaling raw abundance ratios by the sample median, shifting the central tendency of every sample to a common baseline of 0.0 on the $\log_2$-scale. This operation cancels out uniform sample-level loading differences, technical channel-to-channel variations, and global uptake offsets, ensuring that downstream analyses capture intrinsic differences in cargo stoichiometry rather than bulk vesicle internalization.

Following median centering, unsupervised PCA revealed a substantial redistribution of explained variance across the principal components. While uncentered ordination was heavily dominated by a single scale component (87.4\% along PC1), the centered projection rebalanced the variance distribution. The primary axis (PC1) now accounts for 50.5% of the total variance, maintaining robust and clean separation between non-metastatic PC3 replicates (negative coordinates) and metastatic PC3M replicates (positive coordinates). The secondary axis (PC2) explains 20.1% of the variance, resolving biological heterogeneity within the PC3M cohort into two distinct pairs of replicates (R1/R4 and R2/R3).

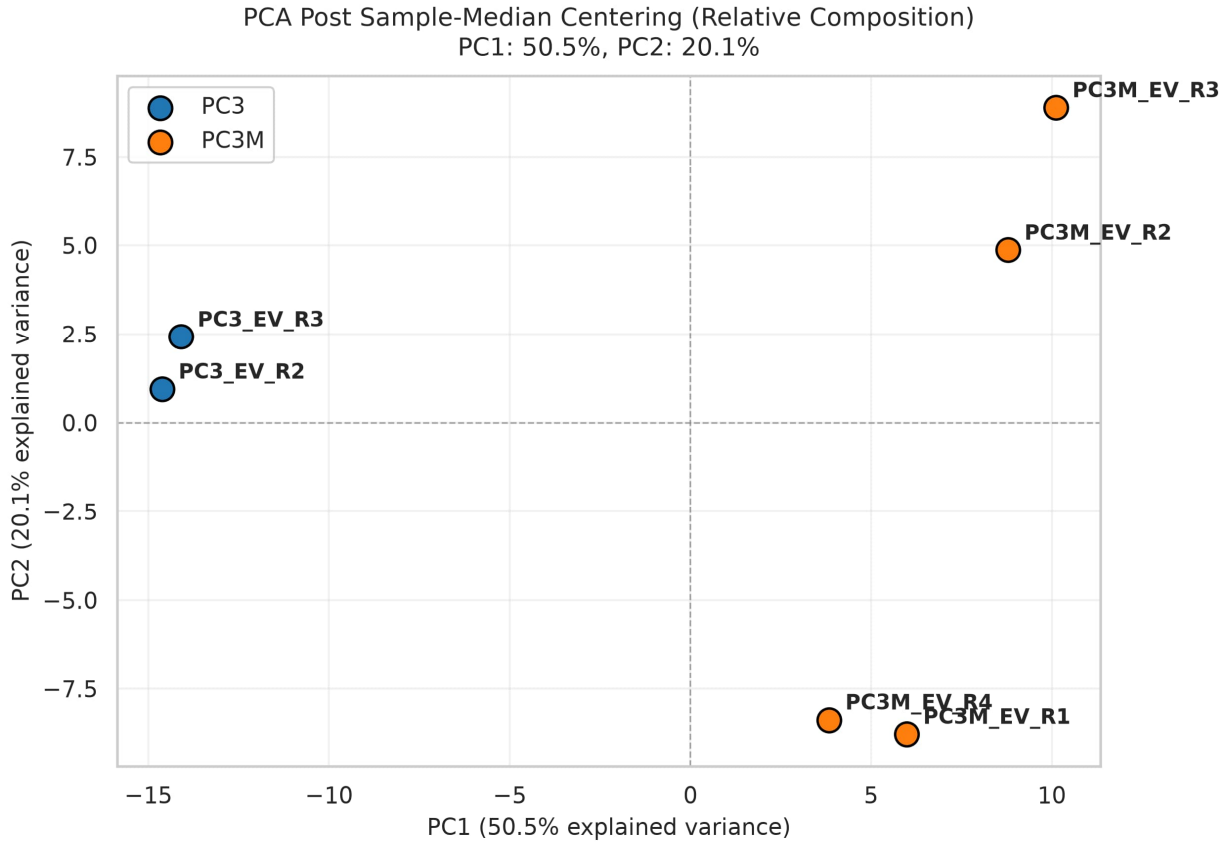


Figure 6. Principal component analysis of relative EV cargo composition following sample-median centering. *Score plot of unsupervised PCA performed on the N = 212 cargo proteins after sample-median centering across the 6 validated biological channels. Subtracting channel medians removes the global scale effect of bulk uptake, reducing PC1 explained variance from 87.4% to 50.5% while preserving distinct separation between non-metastatic PC3 (n = 2, blue) and metastatic PC3M (n = 4, orange) conditions. PC2 captures 20.1% of the variance, delineating intra-group biological structure among metastatic replicates. Individual sample labels correspond to biological replicates within each condition.*

In contrast to the initial uncentered PCA, where 99.5% of proteins displayed positive loadings along PC1, the centered PCA yielded a balanced, bidirectional distribution of feature loadings. Proteins with large positive PC1 loadings and high positive relative fold changes ($\Delta\log_2$) represent cargo selectively enriched in metastatic PC3M uptake. These drivers are dominated by extracellular matrix remodeling enzymes and adhesion molecules, including COL4A2 ($\Delta\log_2$ = +6.01, loading = +0.096), TGM2 ($\Delta\log_2$ = +4.13, loading = +0.095), FN1 ($\Delta\log_2$ = +3.76, loading = +0.095), TNC ($\Delta\log_2$ = +2.11), and ITGA1 ($\Delta\log_2$ = +2.01). Conversely, negative PC1 loadings identify factors preferentially internalized or retained in non-metastatic PC3 uptake, including the classical tetraspanin EV marker CD63 ($\Delta\log_2$ = -1.25, loading = -0.092), signaling kinase MAP3K12 ($\Delta\log_2$ = -2.25, loading = -0.096), stress/ER-associated chaperones HSP90B1 ($\Delta\log_2$ = -1.30) and HYOU1 ($\Delta\log_2$ = -1.22), and cytokine GDF15 ($\Delta\log_2$ = -1.47). This bidirectional structure confirms that sample-median centering successfully decoupled bulk internalization from stoichiometric cargo selectivity.

6. Baseline Differential Uptake Screening via Unmoderated Welch's t-Test

Differential uptake between non-metastatic PC3 (n_1 = 2) and metastatic PC3M (n_2 = 4) conditions was initially evaluated using an unmoderated Welch's two-sample t-test across all N = 212 centered cargo proteins. Unlike standard Student's formulations, Welch's test accommodates unequal sample sizes and heterogeneous variances by calculating effective degrees of freedom via the Welch–Satterthwaite approximation:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

Significance was corrected for multiple hypothesis testing using the Benjamini–Hochberg False Discovery Rate (FDR) procedure, setting baseline discovery thresholds at FDR < 0.10 (q < 0.10) and $|\log_2 FC| \geq 1.0$.

The screening identified **22 differentially abundant cargo proteins** satisfying both significance criteria (14 enriched in PC3M uptake, 8 enriched in PC3 uptake). Proteins with elevated relative uptake in the metastatic condition were primarily associated with cell-matrix interactions and extracellular matrix organization, led by the basement membrane component COL4A2 ($\Delta\log_2$ = +6.01, q = 0.033), junctional catenin CTNND1 ($\Delta\log_2$ = +1.97, q = 0.015), integrin subunits ITGA1 ($\Delta\log_2$ = +2.01, q = 0.037), ITGA3 ($\Delta\log_2$ = +1.54, q = 0.035), and ITGB1 ($\Delta\log_2$ = +1.39, q = 0.083), as well as adhesion/matrix modulators TNC, FN1, and CCN1. Conversely, cargo enriched in PC3 uptake comprised stress-response and endoplasmic reticulum chaperones, including MAP3K12 ($\Delta\log_2$ = -2.25, q = 0.015), signal peptide peptidase HM13 ($\Delta\log_2$ = -1.77, q = 0.090), vesicular cargo receptor LMAN2 ($\Delta\log_2$ = -1.70, q = 0.046), cytokine GDF15 ($\Delta\log_2$ = -1.47, q = 0.046), and ER lumen chaperones HSPA5 and PDIA6.

While Welch's formulation captures genuine large-effect drivers, its unmoderated denominator remains inherently unstable under severe sample size constraints. With $n_1 = 2$, residual degrees of freedom for the PC3 arm collapse to $n_1 - 1 = 1$, restricting total effective degrees of freedom ν to an unstable range ($1 \leq \nu \leq 3$). In this low-degree-of-freedom regime, the standard error in the denominator is dominated by sample variance fluctuations. Consequently, features whose sample variances artificially underestimate true population variance purely by chance achieve deflated p-values and risk false-positive detection. Simultaneously, proteins with biologically genuine effect sizes but slightly elevated sample variance suffer an overly penalizing loss of statistical power under multiplicity correction, leading to widespread false negatives (e.g., failure of key drivers like TGM2 to cross nominal thresholds despite large fold shifts). These limitations necessitate a moderated shrinkage framework that borrows variance information across the entire proteomic ensemble.

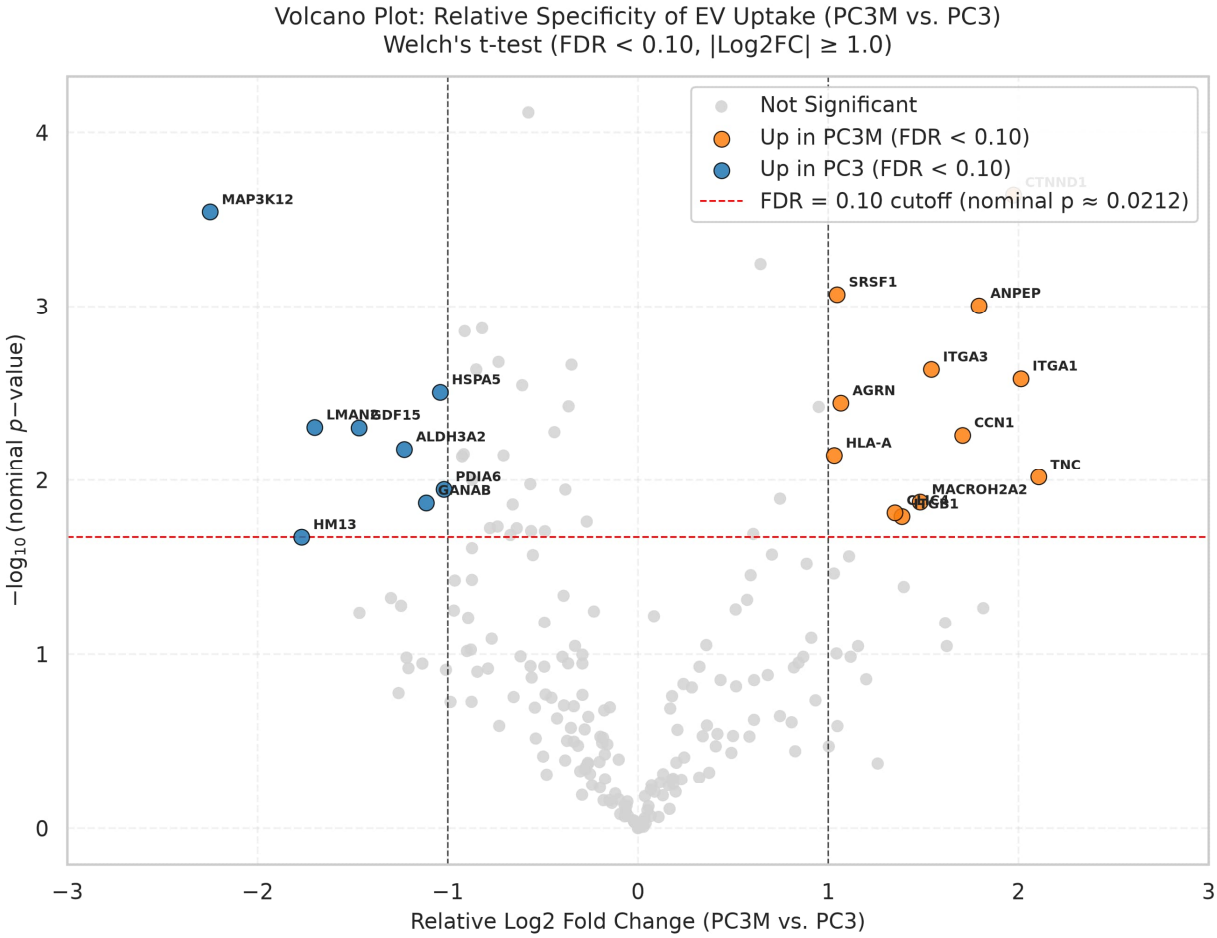


Figure 7. Baseline differential EV cargo uptake screening via unmoderated Welch's t-test. *Volcano plot illustrating relative cargo uptake differences between metastatic PC3M (n = 4) and non-metastatic PC3 (n = 2) conditions across N = 212 validated proteins. The horizontal axis indicates relative effect size (Δlog_2, PC3M vs. PC3), and the vertical axis denotes statistical significance ($-log_{10}$, nominal p-value). Dashed vertical lines demarcate fold-change thresholds ($|log_2 FC| \geq 1.0$), while the dashed red line marks the Benjamini-Hochberg FDR < 0.10 boundary (nominal p ≈ 0.0212). A total of 22 proteins passes dual-cutoff filtering (orange circles: enriched in PC3M, n = 14; blue circles: enriched in PC3, n = 8; grey circles: non-significant, n = 190). Labeled points indicate primary gene symbols.*

7. Empirical Bayes Moderated Differential Uptake Analysis

To overcome the severe loss of statistical power and variance instability inherent to small, unbalanced sample sizes ($n_1 = 2$ vs. $n_2 = 4$, nominal residual degrees of freedom $d_g = 4$), an Empirical Bayes linear modeling framework (limma-trend) was applied across all N = 212 centered cargo proteins. Individual sample standard deviations (s_g) exhibited extreme scatter-dropping toward zero in proteins with artificially concordant replicates or flaring in high-noise features. By modeling prior technical variance s_{0g}^2 as a smooth function of mean abundance (Amean, $\log_2$ scale) and borrowing variance information across the entire proteomic ensemble, posterior standard deviations ($\tilde{s_g}$) were regularized: underestimated sample variances were pulled upward toward the prior trend, while extreme outlier variances were moderated downward. This effectively augmented residual degrees of freedom from $d_g = 4$ to $d_g + d_0 \approx 7.1$, stabilizing the denominator of the moderated t-statistics ($\tilde{t_g}$) without introducing arbitrary variance offsets.

Testing against the moderated t-distribution combined with Benjamini-Hochberg multiple testing correction (FDR < 0.10, $|\log_2 FC| \geq 1.0$) expanded the cohort of statistically significant differential proteins from 22 (in unmoderated Welch's screening) to **37 high-confidence cargo drivers**. Relaxing the variance penalty allowed true biological drivers with substantial fold shifts to be successfully recovered: metastatic cross-linking enzyme TGM2 ($\Delta\log_2$ = +4.13, q = 0.00077) and canonical exosomal tetraspanin CD63 ($\Delta\log_2$ = -1.25, q = 0.0090), both missed by Welch's test due to degree-of-freedom collapse, emerged among the top significant regulators. Overall, the moderated threshold corresponded to an adjusted nominal significance cutoff of $p \approx 0.0397$ at FDR = 0.10.

The Empirical Bayes moderated framework retained all 22 candidate markers identified during initial Welch's testing without dropping a single hit, while recovering an additional 15 statistically robust drivers (+68% expansion in discovery yield) that previously failed multiple-testing thresholds due to residual degree-of-freedom penalties. The resulting 37-protein signature delineates two distinct functional modules: a metastatic uptake program (n = 21, PC3M-enriched) centered on extracellular matrix remodeling, integrin engagement, and junctional stability (e.g., COL4A2, TGM2, FN1, TNC, ITGA1/A3/B1), and a non-metastatic uptake baseline (n = 16, PC3-enriched) dominated by ER chaperones, stress sensors, and classical endosomal markers (e.g., CD63, MAP3K12, HM13, GDF15, HSP90B1).

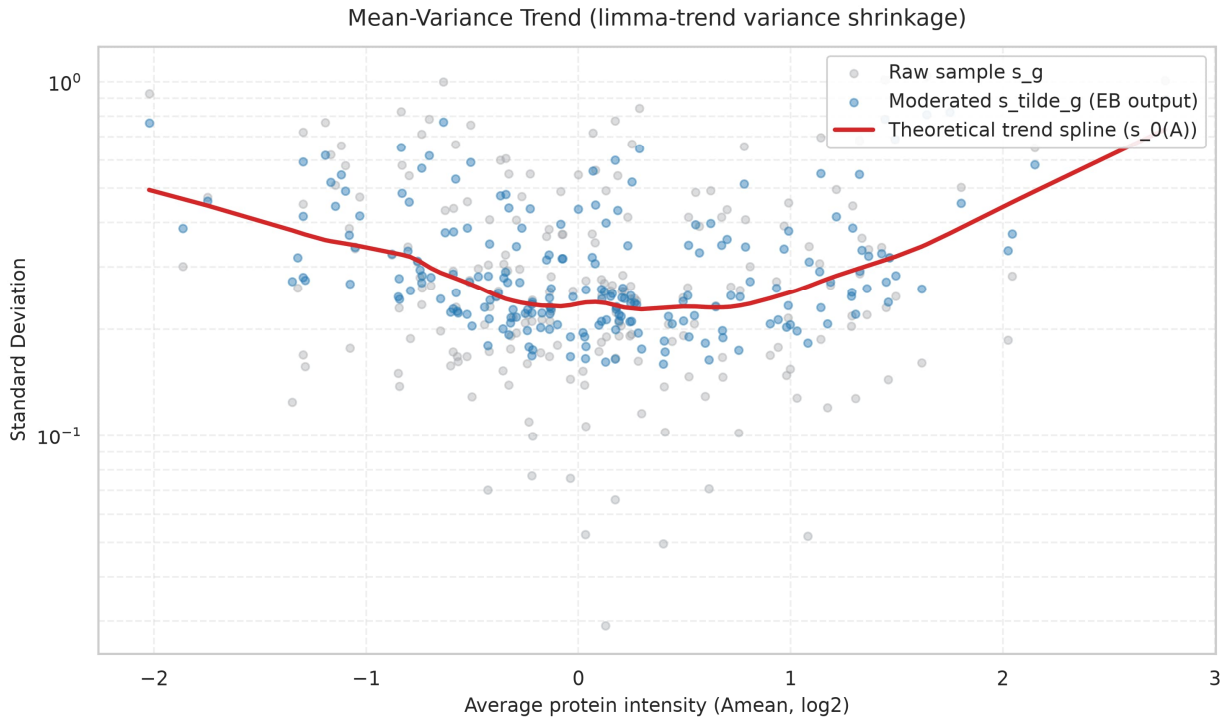


Figure 8. Empirical Bayes variance shrinkage across the EV cargo dynamic range. *Scatter plot displaying the relationship between average protein abundance (Amean, log_2 scale) and residual standard deviation (N = 212). Individual unmoderated sample standard deviations (s_g, grey circles) show marked dispersion, particularly at low sample sizes where stochastic agreement produces artificially deflated variance estimates. The solid red curve denotes the intensity-dependent prior variance trend (s_{0g}(A), modeled via cubic regression splines). Moderated standard deviations ($\tilde{s}_g$, blue circles) represent the posterior estimates obtained following Empirical Bayes moderation (eBayes(..., trend=TRUE, robust=TRUE)), demonstrating continuous shrinkage toward the empirical prior curve.*

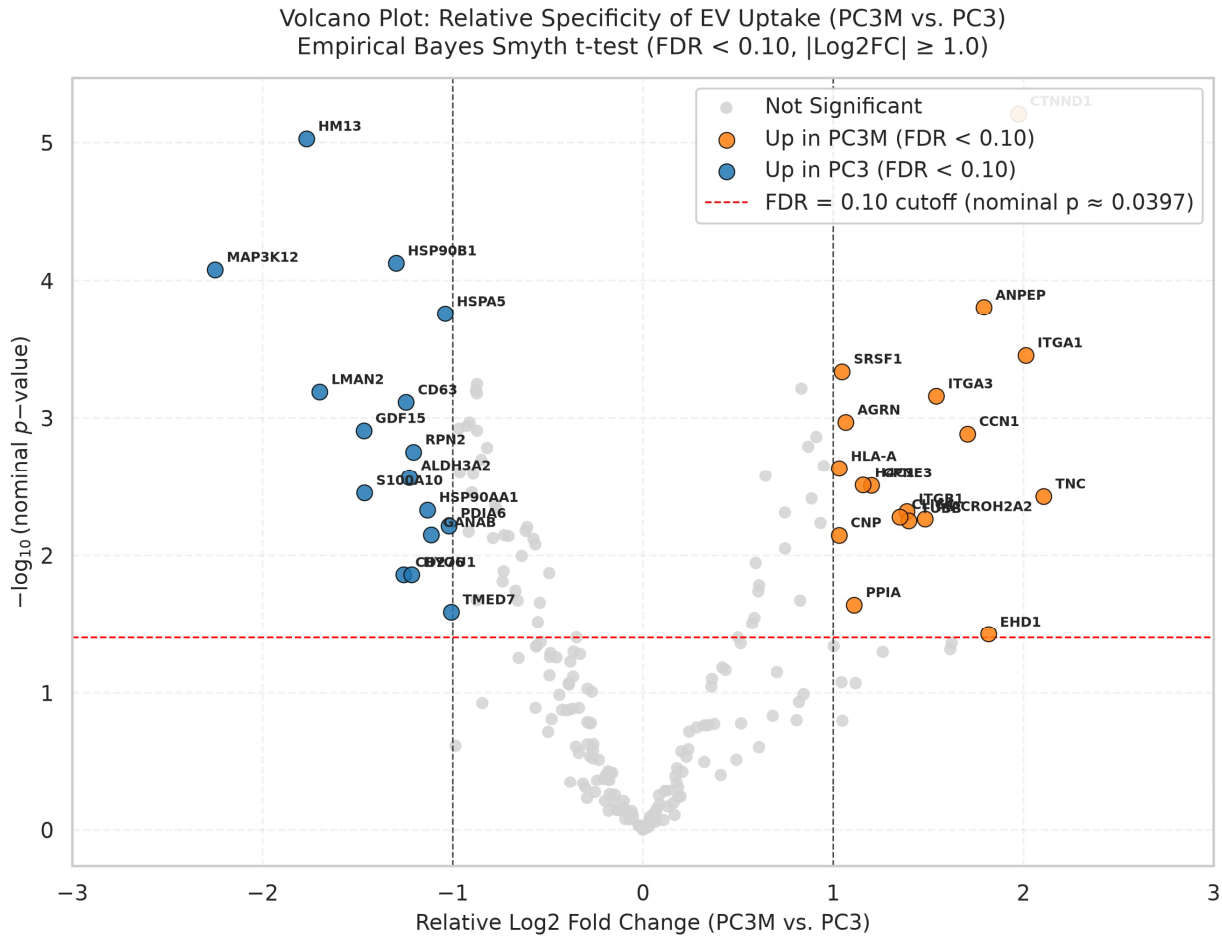


Figure 9. Volcano plot of relative EV cargo uptake evaluated by Empirical Bayes moderated testing. *Differential uptake analysis between metastatic PC3M (n = 4) and non-metastatic PC3 (n = 2) conditions following variance shrinkage. The horizontal axis indicates relative log_2 FC (PC3M vs. PC3), and the vertical axis denotes statistical significance (-log_{10} nominal p-value derived from moderated t-statistics). Dashed vertical lines demarcate fold-change thresholds ($|log_2 FC| \geq 1.0$), and the dashed red horizontal line indicates the multiple testing significance threshold (FDR < 0.10, nominal p ≈ 0.0397). A total of 37 high-confidence proteins passes both criteria: 21 proteins enriched in metastatic PC3M uptake (orange circles) and 16 proteins enriched in non-metastatic PC3 uptake (blue circles); non-significant proteins are displayed in grey (n = 175). Key drivers are labeled by gene symbols.*

8. Unsupervised Hierarchical Clustering and Phenotype Segregation

Unsupervised hierarchical clustering of the 37 empirical Bayes-moderated cargo markers across the 6 biological channels yielded a clean primary bifurcation in the column dendrogram, completely segregating non-metastatic (PC3) from metastatic (PC3M) EV uptake profiles. Because clustering was performed blind to condition labels, this perfect partition confirms that the identified statistical hits capture genuine phenotypic distinctions in EV internalization rather than random analytical noise or batch effects. Within the metastatic arm, the column dendrogram further resolved into two paired sub-clades (R2/R3 and R1/R4), directly recapitulating the intra-group biological structure observed along PC2 in the centered principal component analysis.

The row dendrogram cleanly partitioned the 37 proteins into two mutually exclusive, coregulated expression blocks corresponding to directional uptake tropism:

- **Non-metastatic basal module (n = 16):** Displayed uniform positive Z-scores across PC3 replicates and corresponding suppression across PC3M replicates. This module is composed of endoplasmic reticulum chaperones and proteostasis factors (HSPA5, HSP90B1, HSP90AA1, HYOU1, PDIA6,

GANAB, RPN2), stress-response modulators (MAP3K12, GDF15), vesicular trafficking receptors (LMAN2, TMED7), and the canonical exosomal tetraspanin CD63.

- **Metastatic ECM/invasion module (n = 21):** Displayed pronounced uptake enrichment across all PC3M replicates. This block encompasses structural extracellular matrix components and cross-linking machinery (COL4A2, FN1, TNC, TGM2), integrin heterodimer subunits (ITGA1, ITGA3, ITGB1), junctional stabilizers (CTNND1), matricellular growth factor CCN1, and cell-surface proteases (ANPEP).

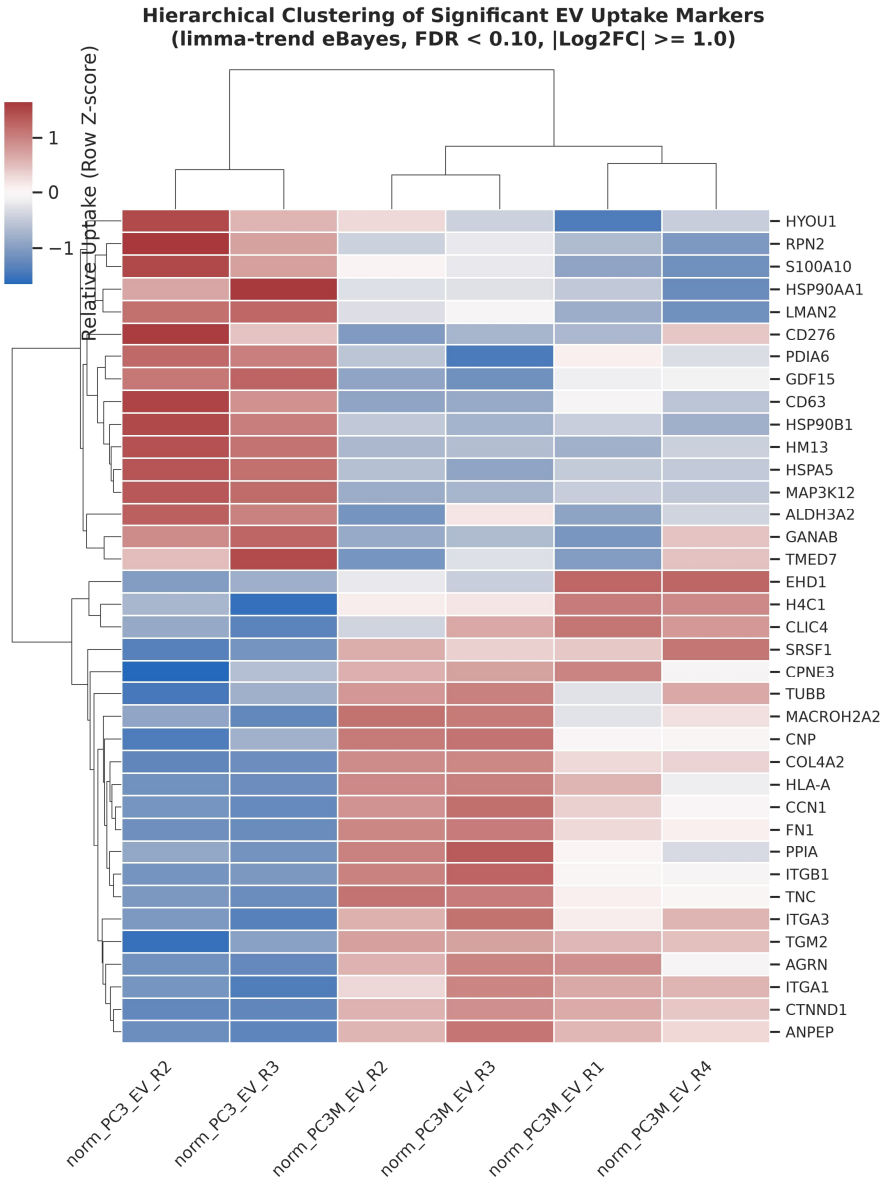


Figure 10. Unsupervised hierarchical clustering of significant EV cargo uptake markers. *Bidirectional heatmap and dendrograms of the 37 differentially internalized proteins identified via Empirical Bayes testing (FDR < 0.10, $|log_2 FC| \geq 1.0$). Columns represent validated biological uptake channels across recipient osteoblasts (SAOS-2) incubated with non-metastatic PC3 (n = 2) or metastatic PC3M (n = 4) EVs. Rows represent individual cargo proteins, standardized via row-wise Z-scores ($z = (x - \mu)/\sigma$; red: relative enrichment, blue: relative depletion). Both rows and columns were clustered using Euclidean distance and average linkage. The column dendrogram exhibits complete, unsupervised bifurcation between PC3 and PC3M conditions, while the row dendrogram separates the cargo proteome into two distinct functional programs: a metastatic ECM/adhesion uptake module (n = 21, bottom) and a non-metastatic proteostatic/endosomal uptake baseline (n = 16, top). Protein rows are labeled with primary gene symbols.*

Concluding Remarks and Future Perspectives

This framework separates cargo selectivity from bulk internalization kinetics in EV uptake proteomics. Through spike-in normalization, background filtering, sample-median centering, and Empirical Bayes variance moderation, the workflow identified 37 cargo drivers separating PC3 and PC3M uptake in osteoblasts. The dataset tracks molecular transitions from proteostasis and endosomal cargo in PC3 to matrix-remodeling and adhesion factors in PC3M.

A key methodological advantage of this framework lies in the mathematical decoupling of stoichiometric cargo selectivity from bulk particle loading uncertainty. While an empirical $2\times$ particle adjustment factor ($\Delta\log_2 = -1.0$) was initially introduced from NTA measurements to approximate absolute uptake scales, the downstream sample-median centering step completely absorbs and cancels any global additive constants across channels. As a result, no parameter fine-tuning, heuristic calibration, or sensitivity screening of vesicle dosing is required: the resulting effect sizes ($\log_2(FC)$), moderated t-statistics, false discovery rates, and ordination topologies (PCA / PLS) reflect invariant, intrinsic differences in vesicular cargo composition rather than experimental dosing variations.

The upstream pipeline restricted analysis to unique peptides to ensure mapping specificity, setting aside 73 multi-mapping features. Future iterations can implement gene-level algorithms to evaluate splice variants and paralogous sequences. Furthermore, assessing internalization differences across signal tiers provides a direction to examine uptake mechanisms. Extending the variance moderation parameters from this model provides a mathematical basis to evaluate rank shifts without low-intensity noise distortions.

Code, matrices, ordination coordinates, and statistical tables are stored in the computational notebook (*main.ipynb*).